THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

December 17, 2018 Time: 9:30 am - 12:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H l; C 12; N 14; 016; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density= 1000 kg/m3
Viscosity (15 °C) = 1.1404 mPa· s
Specific heat= 4.18 kJ/(kg. 0 C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion= 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. Single choice questions (30 marks, 3 marks for each)

1. In most marine waters, nitrogen is the limiting nutrient for the growth of algae, with 0.3 mg/L considered as the critical concentration. At what concentration would phosphorus become the limiting nutrient: (a) 10 mg/L, (b) 1 mg/L, (c) 0.1 mg/L, or (d) 0.01 mg/L.
2. Eutrophication is a nutrient enrichment process of natural waters. Which one of the following statements is false: Eutrophication may result in (a) increased growth of algae, (b) increased water turbidity, (c) increased fish production, and (d) decreased dissolved oxygen during night time on the water surface.
3. Dissolved oxygen, one of the most important water quality parameters, is greatly affected by both eutrophication and stratification. Which statement is false: Stratification in a water body is caused by (a) vertical temperature differences, (b) acidification, (c) vertical salinity differences, or (d) inhibition of vertical mixing.
4. The purpose of water treatment systems is to bring raw water up to drinking water quality. Which one of the following statements is false: A typical water treatment plant in Hong Kong may include the following treatment processes: (a) flocculation, (b) sedimentation, (c) filtration, and (d) desalination.
5. Drinking water quality is defined by physical, chemical and microbiological standards. Which one of the following statements is false: The public of Hong Kong commonly requests the following parameters of drinking water quality: (a) hardness, (b) chloride, (c) iron, and (d) nitrogen.
6. An industrial wastewater (BOD_5 = 1,000 mg/L, total nitrogen = 10 mg/L, flow rate = 1 L/s) is discharged into a public sewer and mixed with domestic sewage (BOD_5 = 200 mg/L, total nitrogen = 40 mg/L, flow rate = 1,000 m^3/d). What is the ratio of BOD_5 to total nitrogen of the mixture: (a) 4:1, (b) 5:1, (c) 6:1, or (d) 7:1.
7. Activated sludge is a (a) physical treatment process, (b) biological treatment process, (c) chemical treatment process, or (d) sludge treatment process.
8. The approximate effluent quality of a conventional secondary wastewater treatment plant has been determined in the laboratory and reported as follows: (a) BOD_5 = 30 mg/L, (b) dissolved solids = 30 mg/L, (c) total nitrogen = 25 mg/L, and (d) total phosphorus = 5 mg/L. Where has the laboratory technician made a mistake?
9. Which compound is NOT considered as a greenhouse gas? (a) CO_2, (b) SO_2, (c) CH_4, (d) O_3
10. Which groundwater remediation technologies is more effective for removing light noaqueorous-phase liquids (LNAPLs)? (a) Pump and treat systems, (b) Soil vapour extraction, (c) In-situ bioremediation, (d) Permeable reactive barriers.

Section B (4 questions with unequal marks)

11. Briefly explain the following terms: **(20 marks, 4 marks** for each one)

    a) Air quality health index (AQHI)
    b) Greenhouse gas
    c) Hydraulic gradient (for groundwater flow)
    d) Basic elements in hydrology cycle
    e) Primary and secondary disinfectant

12. a) Hong Kong's coal-fired Castle Peak Power Station has a gross electricity generating capacity of 4,108 MW. Assume an efficiency of 33.3% with 15% of the waste heat released to the atmosphere as stack heat and the other 85% taken away in the cooling water to the sea. If the cooling water is allowed a rise in temperature by 8.0 ^{0}C, what flow rate would be required, in m^3/s? (Specific heat of water 4,184 J/kg ^{0}C) **(5 marks)**

b) Assume that the 4,108 MW coal-fired power plant, without air pollution control, would generate 900 t/d of sulfur emissions. Assuming 85% efficiency of the sulfur emission control system, how many cubic kilometers of clean air volume will be required daily for dilution in order to meet the Hong Kong's air quality objective of 125 micrograms per cubic meter sulfur dioxide SO_2? **(5 marks)**

13. Given:

Population of Hong Kong in 1950:	2 237,000
Population of Hong Kong in 2000:	6 659,000
Population of Hong Kong in 2016:	7 336,000
Population of Hong Kong in 2066 (baseline population projection):	7 723,000
Population of Hong Kong in 2066 (high population projection):	9 057,000
Population of Hong Kong in 2066 (low population projection):	7 051,000

a) Assuming exponential growth, how much was the average annual net growth rate between the years 1950 and 2000, in % **(3 marks)**

b) Assuming exponential growth, how much would be the average projected annual net growth rate between the years 2016 and 2066 for the baseline population projection and for the high population projection, respectively, in % **(4 marks)**

c) According to The Chief Executive's 2018 Policy Address the Government has decided to commence immediately a study on phased reclamation near Lantau Island for the construction of artificial islands with a total area of about 1 700 hectares and at an estimated cost of 500 billion HKD. The land reserve to be provided by reclamation could accommodate a population of 700 000 to 1 100 000.

(i) Assuming that by 2066 all buildings will be completed, estimate the maximum population density of the reclaimed land, in people per km^2 **(3 marks)**

(ii) Determine the approximate cost of land reclaimed, in HKD per m^2 **(3 marks)**

(iii) List a few possible environmental impacts associated with the reclamation works **(4 marks)**

(iv) Looking at the different population projections for 2066, state which population projection, in your view, might have been used by the Chief Executive. Give a reason why you support or reject the Government's assumption. **(3 marks)**

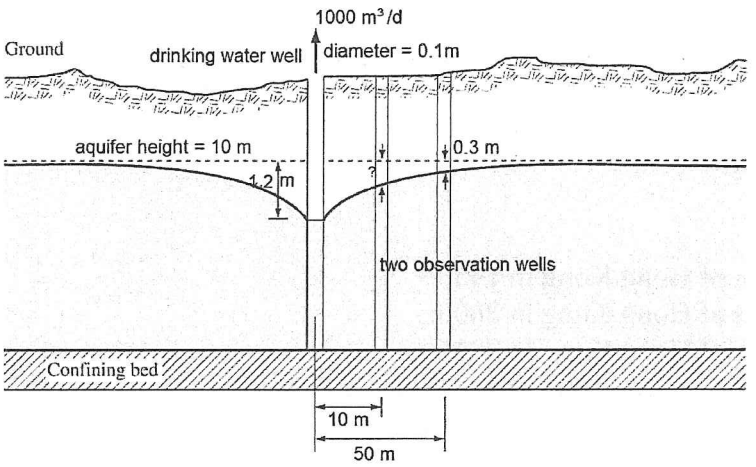
14. Drawdown at a 0.20-m-diameter, drinking water well, which has been pumping at the rate of 1000 m^3/day for a long enough time that steady-state conditions have been reached, is determined to be 1.2 m. The aquifer is unconfined and 10.0 m thick. An observation well 50 m away has been drawn down by 0.30 m.

(i) Determine the hydraulic conductivity of the aquifer. **(5 marks)**

(ii) Estimate the drawdown at the second observation well 10 m away from the production well. **(5 marks)**

(iii) Estimate how long after the observation well 50 m away is contaminated would you expect the drinking water well to be contaminated theoretically? The aquifer has a porosity of 0.35. **(5 marks)**

(iv) Discuss what factors may affect the transport velocity of contaminant in groundwater flow in reality? **(5 marks)**


1000 m³/d
Ground
drinking water well
diameter = 0.1m
aquifer height = 10 m
1.2 m
0.3 m
?
two observation wells
Confining bed
10 m
50 m

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